

CEH v13 Practice Questions – Module 3:

Scanning Networks

1. Which of the following is a common objective of network scanning?

- A. To deliver malware payloads
- B. To identify active hosts and open ports
- C. To perform a brute-force attack
- D. To alter firewall rules

Answer: B

2. Which Nmap scan type performs a TCP three-way handshake?

- A. SYN scan
- B. TCP connect scan
- C. NULL scan
- D. FIN scan

Answer: B

3. Which Nmap option enables OS detection?

- A. -O
- B. -sP
- C. -A
- D. -T4

Answer: A

4. Which port is used by the SSH protocol?

- A. 22
- B. 23
- C. 80
- D. 443

Answer: A

5. What is the purpose of a ping sweep?

- A. To scan for vulnerabilities
- B. To identify active hosts on a network
- C. To bypass firewalls
- D. To brute-force login credentials

Answer: B

6. Which scan is used to determine the state of a TCP port by sending packets with no flags set?

- A. NULL scan
- B. FIN scan
- C. Xmas scan
- D. SYN scan

Answer: A

7. What does a FIN scan send to the target port?

- A. No packet
- B. FIN flag only
- C. SYN-ACK flag
- D. RST flag

Answer: B

8. Which of the following is the default port for HTTPS?

- A. 80
- B. 443
- C. 21
- D. 8080

Answer: B

9. Which command performs a full TCP port scan using Nmap?

- A. `nmap -sS -O`
- B. `nmap -p-`
- C. `nmap -sV`
- D. `nmap -A`

Answer: B

10. Which scanning technique can be used to detect UDP services?

- A. SYN scan
- B. TCP connect scan
- C. UDP scan
- D. ACK scan

Answer: C

11. Which of the following indicates an open TCP port during scanning?

- A. RST response
- B. ICMP unreachable
- C. SYN-ACK response
- D. FIN-ACK response

Answer: C

12. Which tool is used to identify services and versions running on open ports?

- A. Wireshark
- B. Netstat
- C. Nmap with -sV
- D. Tcpdump

Answer: C

13. Which scanning technique is typically used to map firewall rules?

- A. TCP connect scan
- B. SYN scan
- C. ACK scan
- D. FIN scan

Answer: C

14. Which of the following tools can be used to scan a network for live hosts?

- A. theHarvester
- B. ping sweep tool

- C. sqlmap
- D. Burp Suite

Answer: B

15. Which flag combination is used in TCP Xmas scan?

- A. URG, PSH, FIN
- B. SYN, ACK, RST
- C. FIN, ACK, URG
- D. RST, PSH, URG

Answer: A

16. What is the role of an IP ID field in OS fingerprinting?

- A. Identifies which ports are filtered
- B. Tracks session IDs
- C. Helps determine the OS by analyzing how IDs are incremented
- D. Verifies DNS resolution

Answer: C

17. Which scanning method is the least likely to be detected by a firewall?

- A. TCP connect scan
- B. UDP scan
- C. FIN scan
- D. SYN scan

Answer: C

18. What does Nmap's -sS option do?

- A. Conducts a TCP connect scan
- B. Sends SYN packets to perform a stealth scan
- C. Initiates a full connection
- D. Conducts a UDP scan

Answer: B

19. Which scanning phase involves identifying the live systems on a network?

- A. Port scanning
- B. Vulnerability scanning
- C. Host discovery
- D. Enumeration

Answer: C

20. What does a response of ICMP Type 3 Code 13 mean?

- A. Host unreachable
- B. Port unreachable
- C. Communication administratively prohibited
- D. Destination network unknown

Answer: C

21. Which tool is most suitable for a large-scale port scan?

- A. Wireshark
- B. Nikto
- C. Nmap
- D. Hydra

Answer: C

22. What can be used to determine the operating system during scanning?

- A. FTP enumeration
- B. TTL and TCP window size analysis
- C. DNS lookup only
- D. Banner grabbing via ICMP

Answer: B

23. Which scan type sends packets with no flags set to identify open ports?

- A. ACK scan
- B. NULL scan
- C. SYN scan
- D. TCP connect scan

Answer: B

24. What is the effect of scanning a network using a high timing template (-T5)?

- A. Avoids detection
- B. Increases scan reliability
- C. Increases scan speed but may raise alerts
- D. Uses fewer resources on the target

Answer: C

25. Which scan type is best for discovering live hosts without port scanning?

- A. SYN scan
- B. NULL scan
- C. Ping scan (-sn)
- D. Version scan (-sV)

Answer: C